ثانوية المدينة التعليمية
Education City High School

2.14

Logarithmic Function Context and Data Modeling

Core-to-Challenge Worksheet and AP Practice

AP Precalculus - Unit 2 | ECHS Mathematics | Teacher: Mohammad Abu-Ghuwaleh

Student Information

Name: ____________________ Class: ____________ Date: ____________

Directions

Show mathematical evidence. Retain domain restrictions, label graphs, include units, and write complete interpretations. Calculator use is permitted only when the item explicitly requires approximation, regression, or numerical solving.

Readiness

1. Write one precise sentence explaining this principle: A logarithmic model often fits responses that change by equal amounts when input is multiplied by equal factors.

2. Write one precise sentence explaining this principle: The domain must keep the logarithm argument positive.

3. Write one precise sentence explaining this principle: Two data points determine scale and shift once a base and horizontal shift are chosen.

4. Write one precise sentence explaining this principle: Extrapolation toward the vertical asymptote or far beyond data can be especially unstable.

Core

5. Find $y = a + b \ln x$ through $(1, 2)$ and $(e^2, 8)$.

6. Solve $12 = 4 + 2 \ln x$.

7. Give a contextual limitation of a sound-level logarithmic model.

8. State what the formula $y = a + b \ln(x - h)$ tells you and name any required restriction.

9. State what the formula $y = a + b \log_{10}(x - h)$ tells you and name any required restriction.

10. State what the formula $x = h + e^{(y-a)/b}$ tells you and name any required restriction.

Medium

11. Construct or use the logarithmic model: through (1,3),(e,7) in form a+b ln x.

12. Construct or use the logarithmic model: VA x=2, point (3,5), form ln(x-2)+k.

13. Construct or use the logarithmic model: change +6 when x multiplied by 10, f(1)=2.

14. Construct or use the logarithmic model: through (1,0),(100,8) base 10.

15. Construct or use the logarithmic model: model y=5+2ln x solve y=11.

Hard

16. A response increases 3 units whenever input doubles and equals 10 at x=4. Construct a base-2 model.

17. Find inverse prediction for $y = 12 - 4\ln(x+1)$.

18. Explain why a log model may fit diminishing returns.

19. A log regression predicts y=-2 at x=0. Is that necessarily invalid?

Difficult

20. Find a,b in $y = a + b\ln x$ through (2,5),(8,11).

21. A model y=a+b log10 x has y=4 at x=10 and y=10 at x=1000. Find model.

22. Compare extrapolation near x=0 for y=ln x and y=ln(x+1).

Challenge / HOT

23. Prove a logarithmic model linearizes data when plotted against log input, not necessarily semi-log output.

24. Construct a log model with domain x>5, decreasing, concave up, and point (6,2).

AP-Style Multiple Choice

25. Log model domain requires

- **(A)** argument positive
- **(B)** argument negative
- **(C)** x integer

(D) output positive

26. Equal multiplicative input factors produce equal
- **(A)** input differences
- **(B)** output differences
- **(C)** output ratios
- **(D)** zeros

27. Inverse of y=a+b ln x is
- **(A)** $x = \ln((y-a)/b)$
- **(B)** $x = e^{(y-a)/b}$
- **(C)** $x = (y-a)^b$
- **(D)** $x = b/(y-a)$

28. A vertical asymptote limits
- **(A)** output range only
- **(B)** input domain
- **(C)** slope sign only
- **(D)** data count

29. Log models often represent
- **(A)** constant absolute rate
- **(B)** diminishing returns
- **(C)** periodic motion
- **(D)** finite sequences only

30. Changing log base in a model can be absorbed into
- **(A)** vertical coefficient
- **(B)** horizontal shift only
- **(C)** domain deletion
- **(D)** new asymptote always

AP-Style Free Response

31. A skill score is modeled by $S(h) = 20 + 12\ln(h+1)$, h practice hours. (a) Interpret S(0). (b) Find score at h=9. (c) Solve hours for score 50. (d) State two limitations.

32. An intensity scale follows $L = a + b\log_{10} I$. Measurements: L=30 at I=10^-4 and L=70 at I=1. Find model and explain effect of multiplying I by 100.

Reflection

Identify one item where a domain restriction, unit, assumption, or representation changed your reasoning. Explain in 2-3 sentences.

Original ECHS questions. Selected concepts are informed by Pearson and Holt Precalculus; no textbook exercises are reproduced.